

SAKSHAM BHARDWAJ

bhardwajsaksham355@gmail.com
+919811392683
Faridabad, Haryana
[LinkedIn](#) | [Github](#) | [Leetcode](#) | [Web Profile](#)

EDUCATION

B.Tech in Computer Sciences and Bio Sciences

Indraprastha Institute of Information Technology, Delhi (IIITD)
2023 - 2027
CGPA - 8.08 (5 semesters)

Competitive Exam Performance

JEE Mains 2023 All India Rank: 20,565

Senior Secondary Education (XII)

CBSE Board
2023
Percentage: 93.6%

Senior Secondary Education (X)

CBSE Board
2021
Percentage: 96.8%

TECHNICAL SKILLS & KEY COURSES TAKEN

Programming Languages: Java, C, C++, Python

Web Development: React.js, Node.js, RESTful APIs, SQL, PostgreSQL

Tools & Technologies: JavaFX, lib-GDX, Box2D, Streamlit, CrewAI, LangChain, Bio-python, Scikit-learn

Computer Science Fundamentals: Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, Multithreading

Machine Learning: Supervised/Unsupervised Learning, Pattern Recognition, Bayesian Decision Theory, Discriminant Analysis (PCA, LDA, ICA), Ensemble Methods (Bagging, Boosting), Model Evaluation

Bio Sciences:

Cell Biology, Genetics and Molecular Biology, Practical Bio-informatics, Biophysics, Bio-Algorithms

CAREER OBJECTIVE

As a driven and passionate computer science graduate, I am eager to gain practical experience and develop my skills in software engineering. With a strong foundation in programming languages, data structures, and algorithms, I aim to contribute to innovative projects, learn from experienced professionals, and grow my technical and problem-solving abilities.

EXPERIENCE

Undergraduate Researcher at RayLab | IIITD

Conformational Dynamics ATP-Bound vs Unbound proteins via Coarse Grained MD (January 2026 – Present)

- Perform feature engineering and dimensionality reduction on long-timescale coarse-grained MD simulations, cluster conformational states, identify key structural descriptors, and model transition pathways between representative protein conformations.

Deloitte Technology Job Simulation (July 2025)

[Certificate](#)

- Completed job simulation designed by Deloitte, focusing on real-world technology and software engineering scenarios.
- Developed code-based solutions to address business challenges and authored a comprehensive proposal outlining the architecture and implementation plan for a data-driven dashboard.
- Skills Demonstrated:** Python, JSON, data structures & algorithms with Project Planning, Solution Design & Software Development Processes

PROJECTS

ML Library–Simple Machine Learning Library (June 2025 - Present)

[GitHub](#)

- Built and published a Python library implementing core ML algorithms from scratch. Developed a modular library (ml_library) with algorithms including PCA, FDA, LDA/QDA, AdaBoost, Gradient Boosting, Polynomial Regression, and Neural Network utilities. Implemented algorithms purely with NumPy, focusing on mathematical correctness, efficiency, and readability.
- Hosted on GitHub as an open-source educational toolkit, enabling reproducibility and hands-on learning of ML internals.

AI-Powered Trip Planner (May 2025 – June 2025)

[GitHub](#)

- Built an intelligent trip planner using CrewAI, LangChain, and Streamlit to generate customized travel itineraries based on user preferences.
- Orchestrated multiple AI agents and integrated language model operations for dynamic itinerary creation.

Real Estate Management System (January 2025 – May 2025)

[GitHub](#)

- Developed a full-stack property management platform with React.js, Node.js, PostgreSQL, and RESTful APIs.
- Implemented features for property listings, agent tracking, and transaction processing.

Game Development (August 2024 – December 2024)

[GitHub](#)

- Created an Angry Birds clone using lib-GDX and Box2D, implementing game mechanics, physics simulation, and user interactions.

Canteen Management System and Course Management System (August 2024 – December 2024)

- Designed a CLI-based canteen management system in Java with JavaFX.
- Developed a command-line course registration system in Java.

[GitHub](#)

[GitHub](#)

Operating Systems Implementation (August 2024 – December 2024)

- Built a custom shell implementation using system calls and fork operations
- Implemented memory virtualization using segmentation and paging techniques
- Created a multi-threaded system and optimized matrix multiplication.
- Shell: [GitHub](#), Memory: [GitHub](#)

Assembler and Simulator (January 2024 –May 2024)

[GitHub](#)

- Designed an assembler to convert assembly code to binary and a simulator to execute instructions, demonstrating computer organization principles.

Awards and Achievements

- Achieved 98.23 percentile in JEE Mains.
- Robotics: WRO-2017, Robocup-2017/18 Nationals.
- Solved 200+ problems on LeetCode.

Interests and Hobbies

- Robotics and AI.
- Cricket.